

Telehealth Appointment System

1. Objective

To build a Minimum Viable Product (MVP) web application that allows patients to book, view, and attend virtual video consultations with healthcare providers, enabling basic digital healthcare delivery.

2. Target Users

- **Patients:** Individuals seeking to schedule and attend remote medical consultations.
- **Healthcare Providers (Doctors):** Medical professionals who manage their availability and conduct remote consultations.
- **System Administrators:** Staff responsible for managing user accounts and system configuration.

3. Functional Requirements

- **FR-1: Patient Registration and Authentication**
 - **Description:** The system must allow new patients to create an account using an email address and a password of at least 8 characters.
 - **Validation Conditions:** Email must be unique in the database and follow standard format rules (`user@domain.com`). Password must contain at least one number and one uppercase letter.
 - **Role-Based Conditions:** Accessible only by unauthenticated users.
 - **Example Input:** Email: `patient1@example.com`, Password: `Password123`
 - **Example Output:** Account created successfully, activation email sent to `patient1@example.com`.
 - **Edge Case:** Patient attempts to register with an email that already exists; system returns the error "Email already registered" without creating a duplicate record.
- **FR-2: Provider Availability Management**
 - **Description:** The system must allow authenticated providers to set their weekly recurring availability in 30-minute time slots.
 - **Validation Conditions:** Time slots cannot overlap. Start time must be before end time.
 - **Role-Based Conditions:** Accessible only by authenticated users with the `Provider` role.
 - **Example Input:** Date: `2026-06-01`, Start Time: `09:00 AM`, End Time: `05:00 PM`
 - **Example Output:** Availability saved for `2026-06-01` with 16 available slots.

- **Edge Case:** Provider sets an availability slot in the past; system blocks the action with the error "Cannot set availability in the past".
- **FR-3: Appointment Booking**
 - **Description:** Authenticated patients must be able to select a provider and book an open 30-minute availability slot.
 - **Validation Conditions:** The selected slot must currently be marked as available. The patient cannot have another booking at the same time.
 - **Role-Based Conditions:** Accessible only by authenticated users with the **Patient** role.
 - **Example Input:** Provider ID: **101**, Date: **2026-06-01**, Time Slot: **10:00 AM - 10:30 AM**
 - **Example Output:** Booking confirmed, status updated to **Scheduled**, Slot ID status updated to **Booked**.
 - **Edge Case:** Two patients attempt to book the exact same slot simultaneously; the first request processed succeeds, and the second returns the error "Slot no longer available".
- **FR-4: Appointment Cancellation**
 - **Description:** Patients and providers must be able to cancel scheduled appointments up to 24 hours prior to the appointment start time.
 - **Validation Conditions:** Current time must be more than 24 hours before the appointment start time.
 - **Role-Based Conditions:** Accessible by authenticated users with either **Patient** or **Provider** roles linked to that specific appointment.
 - **Example Input:** Appointment ID: **5001**, User: Patient ID **201**
 - **Example Output:** Appointment status changed to **Cancelled**, time slot status reverted to **Available**.
 - **Edge Case:** User tries to cancel 12 hours before the appointment; system returns error "Cancellations are not allowed within 24 hours of the scheduled time".
- **FR-5: Basic Video Consultation Interface**
 - **Description:** The system must provide a secure video consultation room for scheduled appointments that launches when both participants click "Join".
 - **Validation Conditions:** The "Join" button must only activate 10 minutes prior to the scheduled start time and remain active until the scheduled end time.
 - **Role-Based Conditions:** Only the specific patient and provider assigned to the appointment ID can access the room.
 - **Example Input:** Appointment ID: **5001**, Click event on "Join" button at **09:55 AM** for a **10:00 AM** appointment.
 - **Example Output:** Video room connects successfully, displaying local and remote video streams.
 - **Edge Case:** Patient attempts to join the room 30 minutes before the appointment; the "Join" button is disabled and displays a message "Link activates 10 minutes before the start time".

4. Non-Functional Requirements

- **NFR-1: Security (Authentication & Authorization)**
 - All user passwords must be hashed using the bcrypt algorithm before storing them in the database.
- **NFR-2: Performance (Page Load Time)**
 - The patient booking dashboard page must load in under 2.0 seconds when tested on a 4G mobile network connection.
- **NFR-3: Reliability (Uptime)**
 - The core appointment booking service must achieve an uptime of 99.5% over any calendar month.

5. Constraints

- The MVP must run exclusively as a responsive web application supporting the latest versions of Google Chrome, Apple Safari, and Mozilla Firefox.
- The video conferencing capability must be built using open-source WebRTC protocols without custom media server development.

6. Out of Scope

- Integration with external electronic health record (EHR) systems.
- Integration with digital payment gateways for billing.
- Prescription management or pharmacy routing features.